SERVICE 01 / EMBEDDED SYSTEMS

Embedded controls *that behave.*

Firmware built around real hardware, with the states, interfaces and timing needed to make a prototype testable.

FIRMWARE MOTOR CONTROL CAN REAL-TIME SYSTEMS

01 / PROBLEMS THIS ADDRESSES

When the code works *in isolation.*

Embedded behavior becomes difficult to review when state transitions, interrupt timing, communications and hardware assumptions live in different places.

Typical starting points

Unclear operating states, intermittent interfaces, control-loop changes that need hardware evidence, or a prototype that cannot be reproduced consistently.

Firmware architecture

- State machines, timers and interrupt structure
- Interface definitions for sensors, actuators and communications
- PWM, ADC, encoder and control-loop integration
- Fault handling and explicit safe behavior

02 / TECHNICAL CAPABILITIES

Make the control path *reviewable.*

Implementation follows the system boundary: what the MCU owns, what the hardware guarantees and which behavior must be measured at the bench.

03 / ENGAGEMENT WORKFLOW

From interface *to evidence.*

01

Define the boundary

Clarify hardware, timing and command ownership.

02

Inspect the codebase

Trace states, interfaces and build assumptions.

03

Reproduce behavior

Use controlled inputs and observable outputs.

04

Implement and test

Change the smallest useful part, then measure it.

05

Document the handoff

Leave interfaces, tests and next steps clear.

04 / REPRESENTATIVE EVIDENCE

Public work with *hardware behind it.*

PUBLIC TECHNICAL PROJECT

ESP32 Robotics Control Platform

IMU sensing, Bluetooth input, H-bridge PWM, encoder feedback, servo actuation and self-balancing PID control.

VIEW PROJECT   VIEW REPOSITORY ↗

HARDWARE-BACKED DEMONSTRATOR

TI C2000 Actuator Controller

F28069M command handling, state control, ePWM output, setpoint validation, timeout and fault-latched behavior.

VIEW PROJECT   VIEW REPOSITORY ↗

PUBLIC UNIVERSITY RESEARCH

TI Piccolo PMSM dynamometer

Dual-motor control, QEP feedback, SCI-to-CAN migration, real-time testing and validation evidence.

VIEW PROJECT   VIEW REPOSITORY ↗

These are public technical projects, not presented as completed client contracts.

05 / DELIVERABLES

Leave a clearer *next state.*

- Firmware architecture and interface notes
- Prototype code or targeted implementation changes
- State and fault-handling documentation
- Integration and test procedures

> **Scope follows the hardware.**
>
> Deliverables depend on the project stage, available access and the evidence needed for the next decision.

START A TECHNICAL DISCUSSION

Bring the *hard part.*

Describe the project, its current stage, the technical problem and what evidence already exists. Secure file exchange can follow the initial discussion.

START A TECHNICAL DISCUSSION →

Or write directly: info@herdersystemen.com

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